IMPLEMENTATION & DEMO

Course Notes RAG & Adaptive Quiz Engine

A study assistant that answers only from your own lecture PDFs, cites the exact page, and quizzes you on the sections you know least.

BCSE306L Artificial Intelligence · Vellore Institute of Technology

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EVERY ANSWER IS CHECKED

Verified citation
Claim found on the cited page

Auto-cited
Supported sentence linked to its page

Not found
No evidence above τ = 0.35, so no guess

THE PROBLEM

Generic chatbots don't know your lecture notes

Answers you can't check

A general chatbot answers from its training data, not the course's notes, and gives no page to verify against.

Lecture PDFs break naive RAG

Character splitters cut equations in half, scramble table columns, and lose image-only formulas and handwritten scans.

Revision is passive

Re-reading feels productive but gives no feedback on which sections a student actually knows.

> *Practice testing is one of only two learning techniques rated high utility across ages, materials and tasks.*

Dunlosky et al., Psychological Science in the Public Interest, 2013

One workspace: library, chat and adaptive quiz

Library

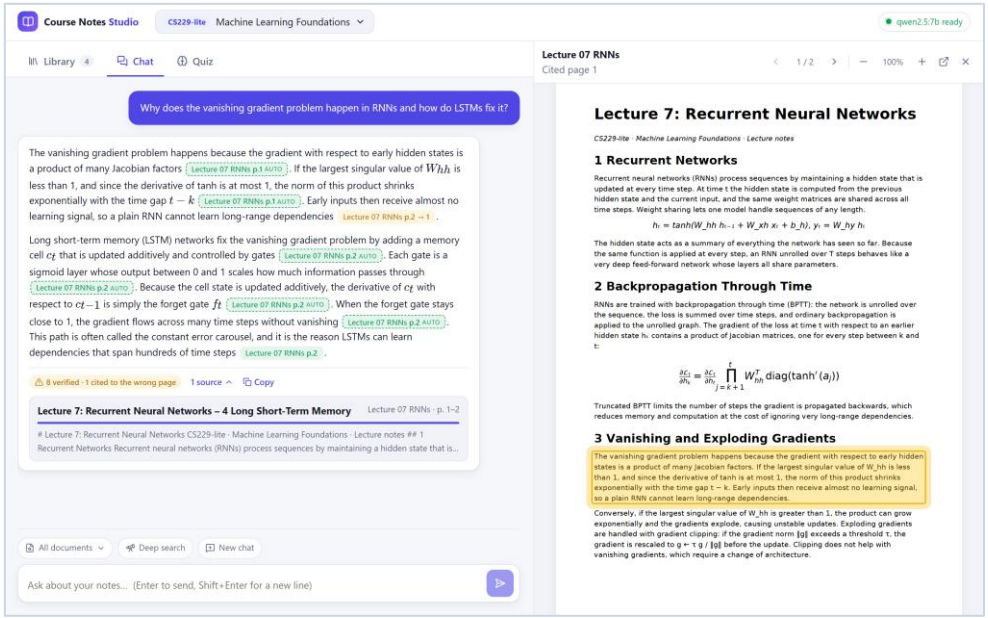
Upload lecture PDFs, Beamer slides or scans. Watch them parse, then preview the extracted Markdown with rendered math.

Grounded chat

Answers stream in with clickable page citations. A click opens the PDF with the supporting paragraph highlighted.

Adaptive quiz

Bloom-calibrated multiple-choice questions with a rationale for every option and a mastery score per section.



Runs fully offline on one laptop: RTX 4070 (8 GB), Qwen 2.5 7B through Ollama.

HOW IT WORKS

Five stages from a lecture PDF to a checked answer

1 Parse

Layout model finds headings, tables and formulas; OCR reads scans; formulas become LaTeX.

Docling · RapidOCR

2 Chunk

Sections become parents (≈800–1200 tokens); 200–300-token children with 50-token overlap are indexed.

Hierarchical chunker

3 Retrieve

Dense and keyword search run in parallel and are fused by rank, then mapped back to parent sections.

BGE-M3 · bm25s · RRF

4 Rerank

A cross-encoder scores each section against the question; nothing above τ = 0.35 means “Not found”.

bge-reranker-v2-m3

5 Verify

Qwen writes a cited answer; every citation is then checked against the retrieved pages.

Qwen 2.5 7B · Ollama

Python · FastAPI · Qdrant · SQLite · React + Vite · server-sent events for streaming

Parsing keeps equations, tables and scans intact

Image-only equation → LaTeX

Bayes' rule recovered as LaTeX from a slide where the formula is only a picture

Table → Markdown rows

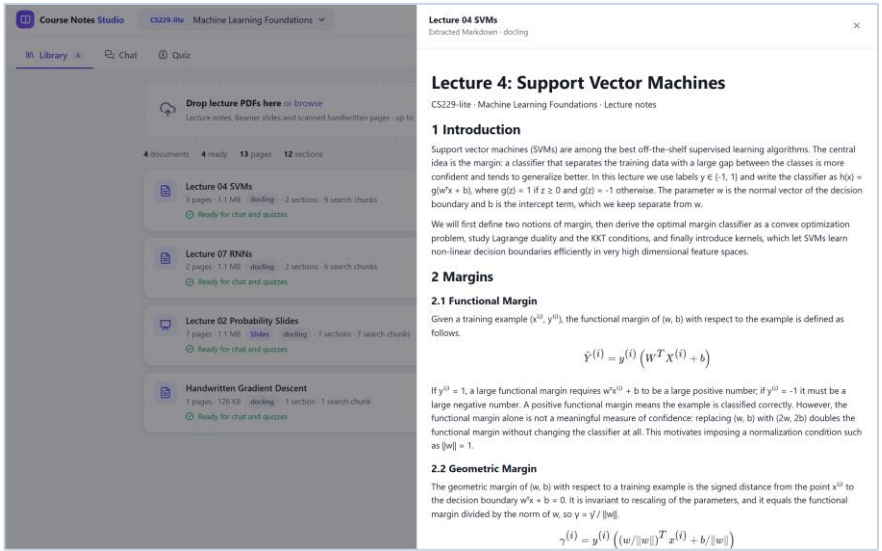
Kernel table keeps its 4 rows × 3 columns; rows are also rewritten as sentences for search

Handwriting → text

Scanned notes are OCR'd: “If alpha is too large, gradient descent can overshoot…”

Structure → parent/child chunks

Headings and slides set the parents; a formula or table is never split



13 pages

41 s on the GPU

23 search chunks

RETRIEVAL

Hybrid search finds both meaning and exact symbols

Dense · BGE-M3
top 25 by meaning

Keyword · BM25
top 25; “α” = “\alpha”, “Lemma 3.2”

Reciprocal Rank Fusion

$$RRF(d) = \Sigma_m \; 1 \,/\, (60 + rank_m(d))$$

top 15 children → their parent sections, so the model sees whole derivations

Cross-encoder rerank
keep top 5 with score > 0.35

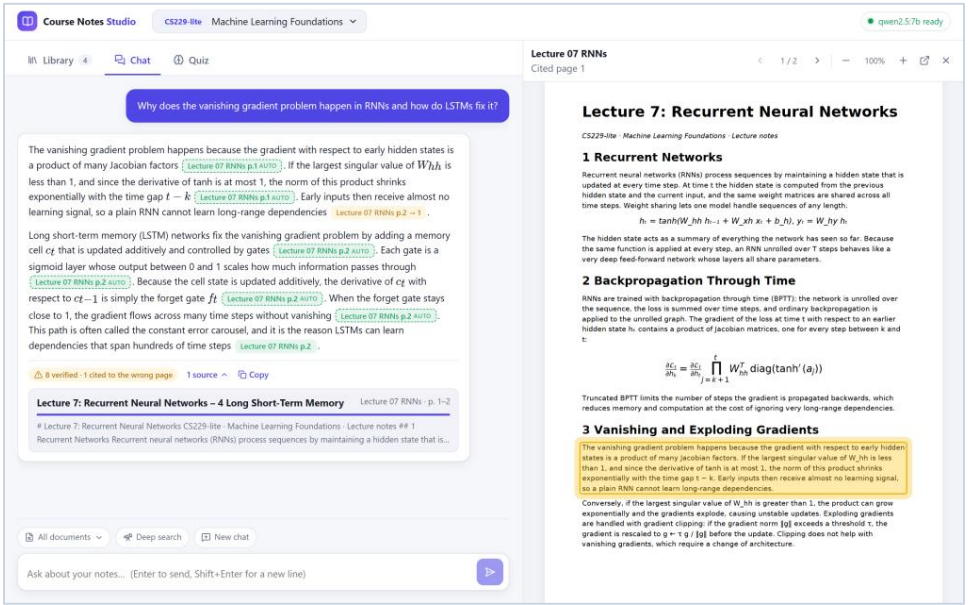
Nothing passes?
reply “Not found in course notes”

Why both?

Embeddings understand paraphrases (“why do early inputs stop learning?”) but blur exact tokens; BM25 with a math-aware tokenizer catches theorem numbers, variable names and Greek letters. Fusing ranks, not scores, needs no tuning between the two.

GROUNDED CHAT

Every citation is checked against the retrieved pages



1 **Source**
the cited file and page must be in the retrieved context

2 **Page**
if the claim's words are on another page, the chip shows p.2 → 1

3 **Highlight**
the viewer opens on the single paragraph that matches the claim

4 **Completion**
uncited but supported sentences get an auto citation; the rest are flagged as unsupported

Sentences with a verified citation (Qwen 2.5 7B): 29% → 100%

Every question is repaired and validated first

Pick sections

70% weighted toward weak sections, 30% by relevance

Number the evidence

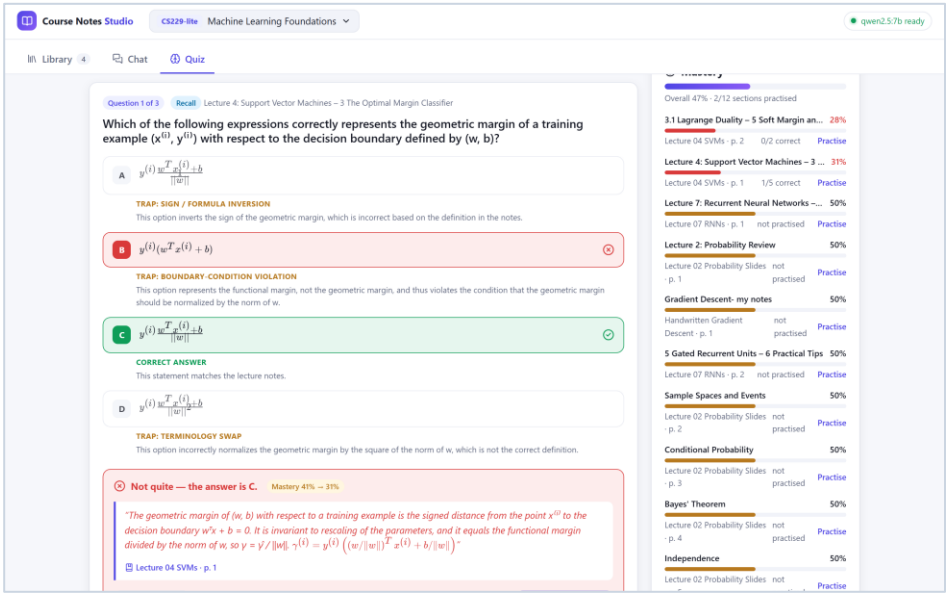
notes shown as [S1], [S2]… so Qwen cites sentence ids, not a retyped quote

Generate to a schema

4 options; each distractor labelled sign inversion, boundary violation or terminology swap

Repair and validate

fix label slips; reject if not exactly one correct, bad LaTeX, or unsupported evidence



Questions passing validation: 33% with free-text quotes → 67% with evidence ids → 100% (9 of 9) with repair

ADAPTIVE PRACTICE

Weak sections come back far more often

After each first attempt

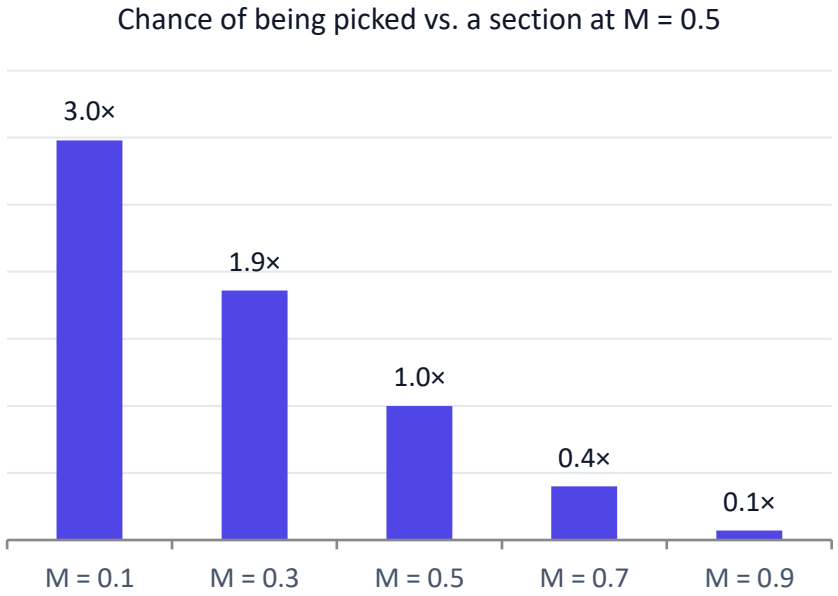
$$M \leftarrow clip(M + \alpha(S - M), 0, 1)$$

S = 1 if correct, 0 if not · α = 0.25 · M starts at 0.5

Choosing the next quiz's sections

$$P(\text{section } c) \propto (1.05 - M_c)^2$$

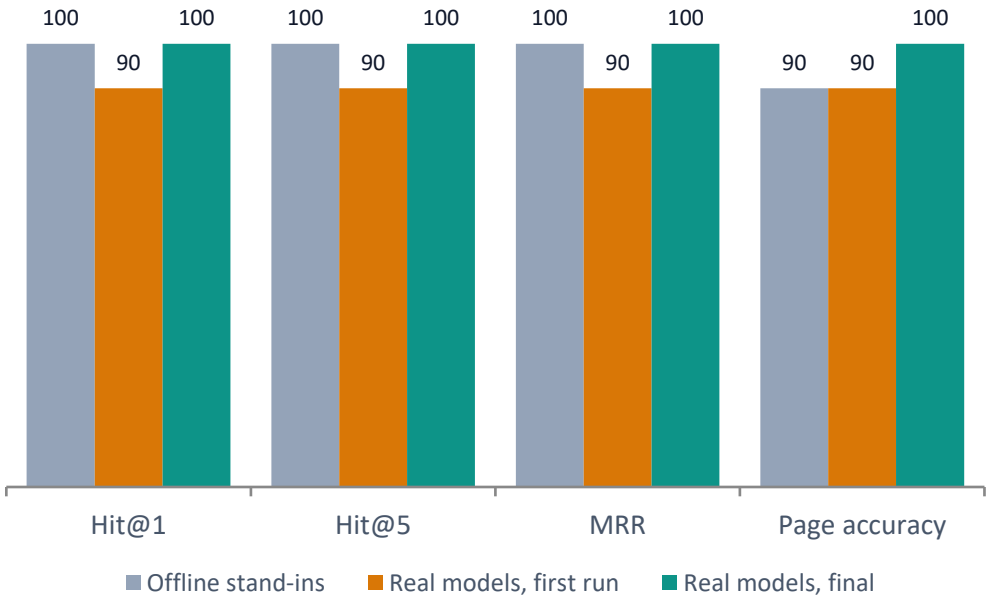
for 70% of slots; the other 30% keep coverage



A section at 10% mastery is picked about 40× more often than one at 90%.

RESULTS

Right lecture section found for 10 of 10 questions



Measure	Final
Hit@1 / Hit@5	100% / 100%
Mean reciprocal rank	1.00
Cited page inside section	100%
Warm query latency	268 ms

The one miss in the first run was “variance of a Bernoulli variable”, answered by a table. Rewriting table rows as sentences for embedding and reranking fixed it (score 0.38 > τ), and lifted the kernel-table question from 0.61 to 0.86.

10 hand-written questions over 4 lecture PDFs (13 pages): paraphrases, theorem numbers, a table lookup and a handwritten scan.

RESULTS

What made a 7B model reliable enough

Sentences with a verified citation

rule repeated in the user turn + server-side completion

29% → **100%**

Quiz questions passing validation

numbered evidence sentences + repair of label slips

33% → **100%**

Benchmark questions found (Hit@1)

table rows rewritten as sentences for search

90% → **100%**

AUTOMATED TESTS, ALL PASSING

289	backend unit + API end-to-end
56	frontend unit (Vitest)
9	browser end-to-end (desktop + mobile)
5	live tests on the real models

Positive and negative cases for every component: invalid, empty, encrypted and duplicate PDFs, off-topic questions, malformed model output.

RELATION TO THE SYLLABUS

Modules 1–4: agents, search, local search, logic

Module	Syllabus topics	Where it appears in this project	Link
M1 Introduction & Intelligent Agents	Agent structure, environments, applications of AI	The app is a learning agent: it perceives notes (parser, OCR) and questions, acts by answering and quizzing, and keeps a student model. The environment is partially observable (knowledge is hidden) and sequential.	**Strong**
M2 Search	State-space search, best-first, uniform cost, A*	Retrieval is search over a space of passages: cosine and BM25 scores order the frontier, RRF fuses the rankings, and best-first pruning keeps 25 → 15 → 5 for the costly cross-encoder.	**Strong**
M3 Local & Adversarial Search	Hill climbing, simulated annealing, minimax	Quiz generation is generate-and-test: score a draft against constraints, repair it, or restart with a higher temperature. Distractors are chosen as the most tempting wrong answers.	**Partial**
M4 Logic & Reasoning	Propositional and first-order logic, forward chaining	Citation and quiz validation are rule-based inference: facts (file, page, section range) fire rules that conclude valid / moved / unsupported; “exactly one correct option” is a propositional constraint.	**Strong**

Modules 5–8: uncertainty, planning, language

Module	Syllabus topics	Where it appears in this project	Link
M5 Uncertain Knowledge	Quantifying uncertainty, Bayes rule, Bayesian networks	BM25 comes from the probabilistic relevance model; the reranker outputs P(relevant | question, passage) and τ = 0.35 is a decision threshold; mastery is a belief about the student, updated by evidence (a simplified knowledge-tracing model).	**Strong**
M6 Planning	Classical, hierarchical planning, acting under uncertainty	A quiz is planned top-down: goal (n questions on weak spots) → slots → section → Bloom level → generate → validate. Ingestion is a tracked plan (queued → parsing → chunking → indexing → ready) with a fallback parser.	**Partial**
M7 Communicating, Perceiving, Acting	Probabilistic language processing, information retrieval, information extraction, perception, object recognition	The core of the project: hybrid retrieval and reranking (IR), layout parsing, tables and formula-to-LaTeX (IE), LLM generation and tokenization (language), layout detection and OCR on page images (perception).	**Strong**
M8 Contemporary Issues	Current topics in AI	Hallucination and grounded generation, verifiable citations, local and private inference on a laptop GPU, academic integrity.	**Strong**

RELATION TO THE SYLLABUS

Course outcomes met by the project

CO1 **Evaluate AI methods and their foundations**

10-question retrieval benchmark (Hit@k, MRR); offline stand-ins vs. real models; before/after ablations of each fix.

CO2 **Apply AI to inference, perception, representation and learning**

Perception (layout, OCR), knowledge representation (parent/child chunks, vectors), inference (verification rules), learning (mastery model).

CO3 **Reason with uncertainty and knowledge representation**

Probabilistic ranking, a relevance threshold that answers “Not found” instead of guessing, mastery as an updated belief.

CO4 **Show how search drives problem solving**

Ranked vector and BM25 search, rank fusion and top-k pruning decide what the model is allowed to say.

WHAT WE LEARNED

Problems we hit, and how we fixed them

Quotes were paraphrased

Qwen rewrote the “exact quote”, so 2 of 3 questions failed. Showing the notes as numbered sentences and asking for ids makes quotes verbatim by construction.

One citation per paragraph

Only 29% of sentences carried a citation. Repeating the rule in the user turn and completing citations on the server reached 100%.

Tables were invisible to search

Pipes and dashes don't look like a question. Rewriting rows as “Distribution: Bernoulli; Variance: p(1 − p).” fixed the benchmark miss.

Every heading was level 1

Docling flattens heading levels, so sections never merged into 800-token parents. Merging now stops only at “Lecture N” / “Chapter N”.

LIMITATIONS & NEXT STEPS

What it can't do yet, and where it goes next

Bigger evaluation

A larger, independently written question set per course, and quiz-quality ratings from students and faculty.

Stronger verification

Use an entailment (NLI) model next to the lexical check, so paraphrased claims are verified too.

Better student model

Replace the moving-average mastery with Bayesian knowledge tracing and spaced-repetition scheduling.

KNOWN LIMITATIONS

- Evaluation scale: 10 benchmark questions over 4 synthetic lecture PDFs; results need confirming on real course material.
- A 7B model still varies run to run: answers may cite few sentences (completion fills the gap), and two distractors can look nearly identical once rendered.
- Formula OCR can misread symbols (ŷ read as Ŷ), and the lexical page check cannot verify paraphrased claims.

THANK YOU

Questions and feedback welcome

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github.com/RoyceRRNS/AI_Project

